

LECTURE 7

* Assignment 2 Group 7 Nov
* Individual assignment 15 Nov
The IS-MPR-PC Model (optional)

Cramer's rule
THE IS CURVE

1. An IS curve relates output to interest rates.
2. The higher interest rates are, the lower output is.
3. It should be noted that the IS relationship is between output and real interest rates, not nominal rates.
4. Real interest rates adjust the headline (nominal) interest rate by subtracting off inflation

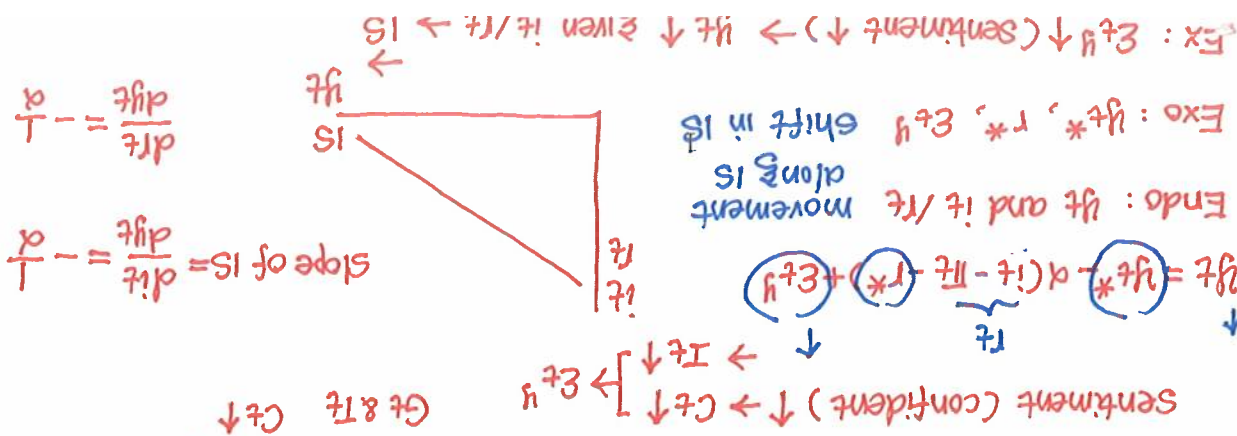
Note: $i_t = r_t + \pi_t$
i.e.

- (1) where r_t is the real interest rate, i_t is the nominal interest rate and π_t is the rate of inflation.
- (2) The IS curve has the following form:

$$y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + e_t^y$$

α - sensitivity of y_t to r_t
 $y_t - y_t^* = -\alpha(i_t - \pi_t - r^*) + e_t^y$
 Expressed in words, this equation states that the gap between output and its natural rate $(y_t - y_t^*)$ depends on two factors:

- The real interest rate $i_t - \pi_t$: The coefficient α describes the effect of a one point increase in the real interest rate on output. We can see from the equation that if $e_t^y = 0$ then a real interest rate of r^* will imply $y_t = y_t^*$. Goods mkt / Money mkt } MPR
- Aggregate demand shocks, e_t^y : Many other factors beyond the real interest rate influence aggregate spending decisions. These include fiscal policy, asset prices and consumer and business sentiment. We will model all of these factors as temporary deviations from zero of an aggregate demand shock, e_t^y . Asset prices $\downarrow \rightarrow$ Richer $\rightarrow$ Gt & r_t $\downarrow$ $C_t \downarrow$



A MONETARY POLICY RULE

- ② $\pi_t \downarrow \rightarrow i_t \downarrow$ b/c CB misses interpret $\rightarrow r_t \downarrow \rightarrow CB \downarrow \rightarrow i_t \downarrow \rightarrow y_t \downarrow$
explosive
(3) $\pi^* = 0\%$ } $i_t \downarrow$
 $\pi_t = 3\%$

:l̥əpɔw mo

$$(*y - y)y + *y + *x = y$$

↑↑! ← ↓↑↑! not why ↓↑! ← ↓↑↑!

2. The central bank adjusts its interest rate in line with inflation with the goal of meeting an inflation target.
- ① $\pi \uparrow \rightarrow i = r + \pi$, it $\uparrow$ b/c MPR that $\beta \pi > 0 \rightarrow r \uparrow$

3. Specifically, the central bank adjusts the nominal interest rate, i_t , upwards when inflation, π_t , goes up and downwards when inflation goes down (we are assuming that $\beta_\pi > 0$) and it does so in a way that means when inflation equals a target level, π^* , chosen by the central bank, real interest rates will be equal to their natural level.

$$PC: \pi_e = \pi_e^* + \lambda(y_e - y_e^*) + \varepsilon_e \pi^*$$

4. Why a rule of this form might be a good idea?
5. Suppose the central bank has a target inflation rate of π^* that it wants to achieve. Ideally, it would like the public to understand that this is the likely inflation rate that will occur, so that $\pi_t^e = \pi^*$. If that can be achieved, then the Phillips curve tells us that, on average, inflation will equal π^* provided we have $y_t = y_t^*$. And the IS curve tells us that, on average, we will have $y_t = y_t^*$ when $i_t - \pi_t = r^*$. So this is a policy that can help to enforce an average inflation rate equal to the central bank's desired target, provided the public adjusts its inflationary expectations accordingly.
- Inflation targeting π^***
- $\pi_t^e = \pi^* \Rightarrow y_t = y_t^*$**

* $\pi^* = \pi^* \Leftrightarrow \pi^* = \pi^*$
inflation targeting π^*

THE IS-MPR CURVE

1. This is the model. It consists of three equations: (a) the Phillips curve, (b) the IS curve and (c) the Monetary Policy Rule.

2. Specifically,

(a) The Phillips curve:

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \epsilon_t^\pi \quad (4)$$

(b) The IS curve:

$$y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \epsilon_t^y \quad (5)$$

(c) The Monetary Policy Rule.

$$i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) \quad (6)$$

3. What about a graphical representation of this model?

IS & MPR $\pi_t \downarrow \rightarrow i_t \downarrow$ b/c $\beta_\pi > 0 \rightarrow r_t \downarrow \rightarrow \text{COB} \downarrow \rightarrow \text{CE} \& \text{IE} \uparrow \rightarrow y_t \uparrow$ IS-MPR downward

4. This is a system of three variables which makes it hard to express on a graph with two axes.

5. To make the model easier to analyse using graphs, we are going to reduce it down to a system with two main variables (inflation and output).

6. We can do this because the monetary policy rule makes interest rates a function of inflation, so we can substitute this rule into the IS curve to get a new relationship between output and inflation that we will call the IS-MPR curve.

7. Substituting the MPR equation (10) into the IS equation (9) gives:

$$y_t = y_t^* - \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \epsilon_t^y \quad (7)$$

8. This is the IS-MPR curve. It combines the information in the IS curve and the MPR curve into one relationship.

9. What would this curve look like?

10. It turns out it depends especially on the value of β_π , the parameter that describes how the central bank reacts to inflation.

11. The IS-MPR curve says that an extra unit of inflation implies a change of $-\alpha(\beta_\pi - 1)$ in output.

12. Is this positive or negative?

13. We are assuming that $\alpha > 0$ so this combined coefficient will be negative if $\beta_\pi - 1 > 0$.

14. In other words, the IS-MPR curve will have a negative slope provided the central bank reacts to inflation so that $\beta_\pi > 1$.

15. The explanation for this is as follows: An increase in inflation of x will lead to an increase in nominal interest rates of $\beta_\pi x$ so real interest rates change by $(\beta_\pi - 1)x$. This means that if $\beta_\pi > 1$, an increase in inflation leads to higher real interest rates and, via the IS curve relation, to lower output. So if $\beta_\pi > 1$ then the IS-MPR curve can be depicted as a downward-sloping curve.

16. In contrast, if $\beta_\pi < 1$, then an increase in inflation leads to lower real interest rates and higher output, implying an upward-sloping IS-MPR curve.

17. For now, we assume that $\beta_\pi > 1$ so that we have a downward sloping IS-MPR curve.

[Figure 7: The IS-MPR curve with $\epsilon_y^e = 0$]

$$\begin{aligned} \text{Fisher Eq} \quad \Delta i_t &= \Delta r_t^* + \Delta \pi_t \\ \text{MPR} \quad \Delta i_t &= \Delta r_t^* + \Delta \pi_t^* + \beta_\pi (\Delta \pi_t - \Delta \pi_t^*) \end{aligned}$$

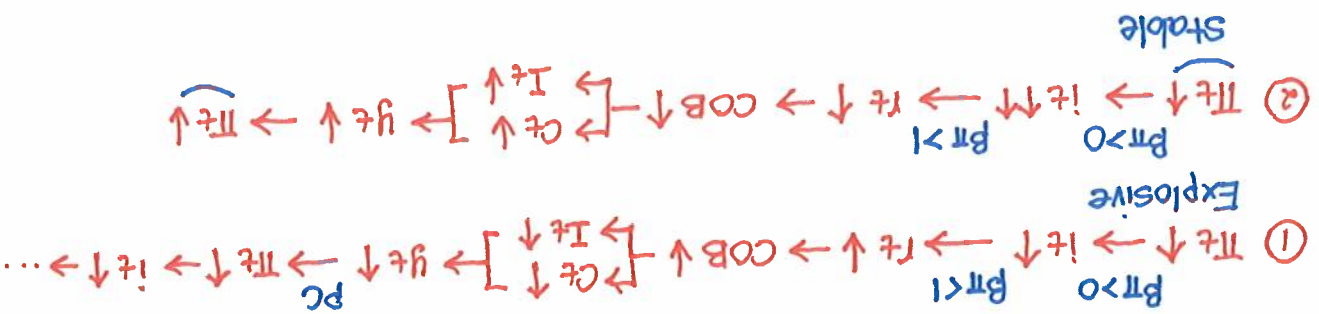
$$\Delta i_t = \beta_\pi \cdot \Delta \pi_t$$

$$\beta_\pi \cdot \Delta \pi_t = \Delta r_t + \Delta \pi_t$$

$$\Delta r_t = (\beta_\pi - 1) \cdot \Delta \pi_t$$

if $\pi_t \uparrow \rightarrow r_t \uparrow$ iff $\beta_\pi > 1$

$$\begin{aligned} \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ i_t = r_t + \pi_t \quad \therefore \quad r_t = i_t - \pi_t \end{aligned}$$



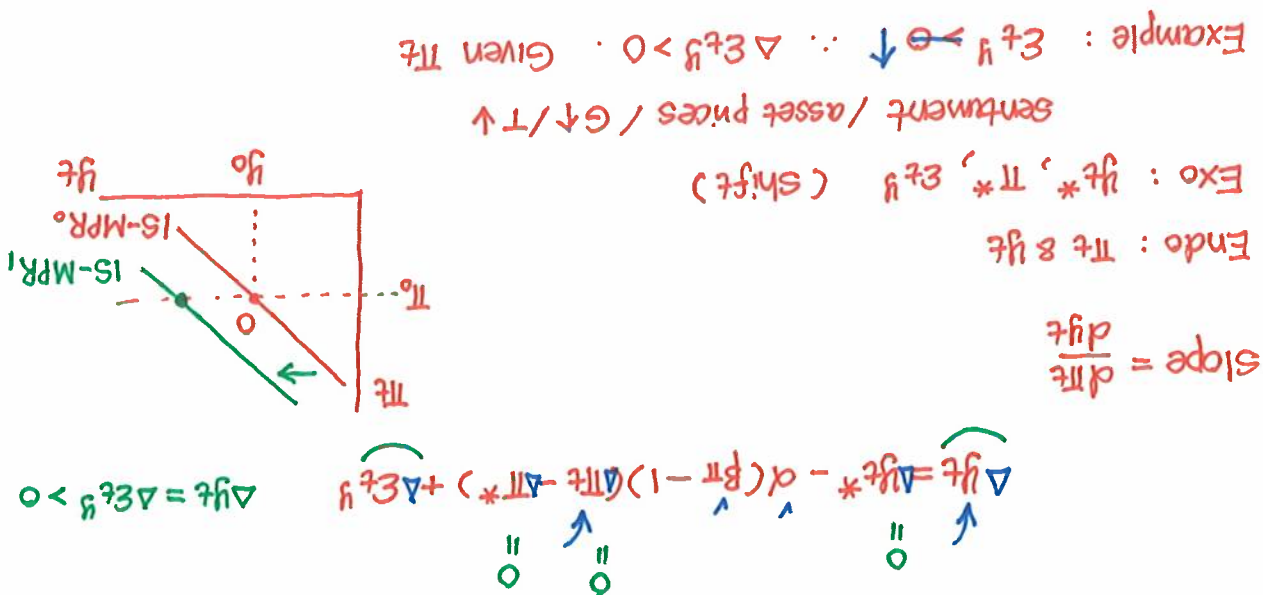
18. Note that when $\epsilon_t^y > 0$, the economy experiences an aggregate demand shock and the IS-MPR curve will shift to the right.

[Figure 8: The IS-MPR curve with $\epsilon_t^y > 0$]

19. Figure 7 thus shows what the IS-MPR curve looks like when the aggregate demand shock $\epsilon_t^y = 0$.

20. Again, the key thing to notice is the value of inflation that occurs when output equals its natural rate. When $y_t = y_t^*$ we get $\pi_t = \pi^*$.

21. Figure 8 shows how the IS-MPR curve shifts to the right if $\epsilon_t^y > 0$ corresponding to a positive shock to aggregate demand.



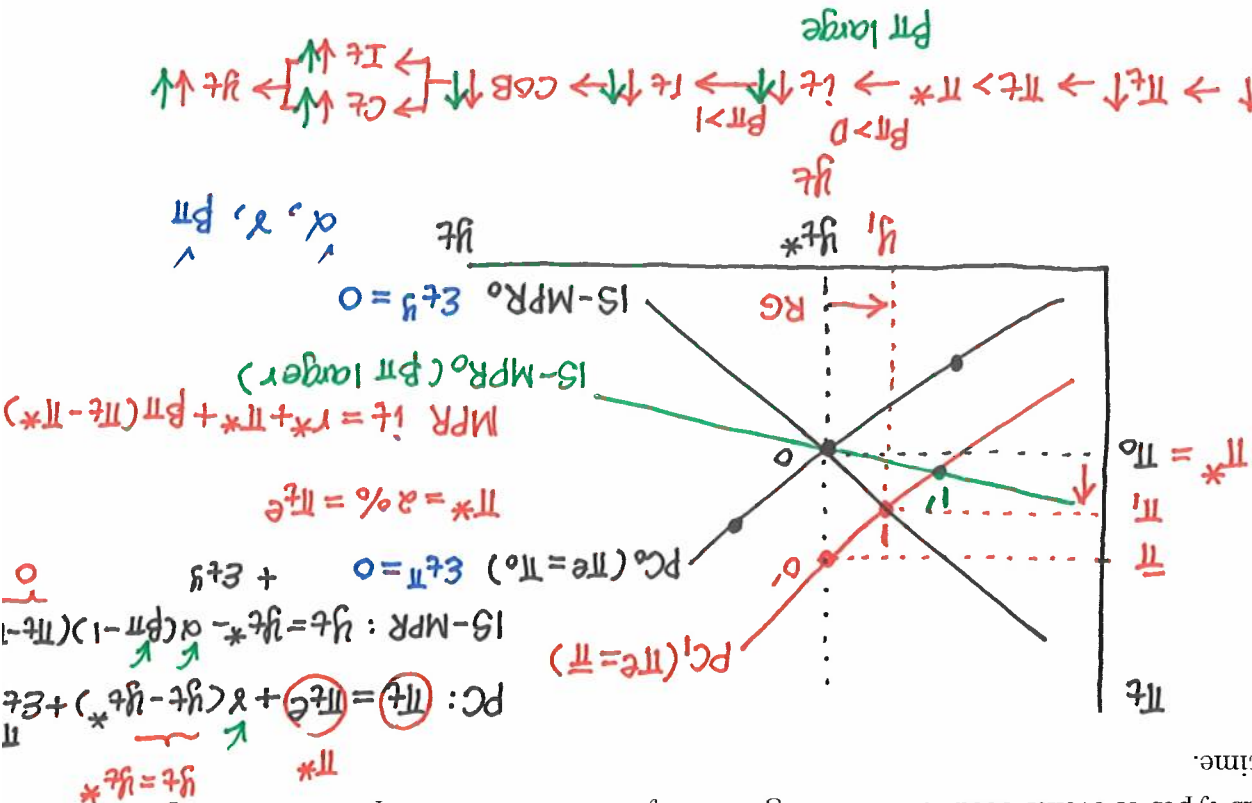
THE IS-MPR-PC MODEL

1. The full IS-MPR-PC model can be illustrated in the traditional fashion as a graph with one curve that slopes upwards (the Phillips curve) and one that slopes downwards (the IS-MPR curve provided that we assume $\beta_\pi > 1$).

[Figure 9: The IS-MPR-PC curve with $\epsilon_t^\pi = 0$ and $\epsilon_t^y = 0$]

2. Figure 9 provides the simplest possible example of the graph. This is the case where both the temporary shocks, ϵ_t^π and ϵ_t^y equal zero and the public's expectation of inflation is equal to the central bank's inflation target.

3. In the next lecture, we will analyse this model in depth, examining what happens when various types of events occur and focusing carefully on how inflation expectations change over time.



Recall the slope of IS-MPR = $-\frac{1}{\alpha(\beta_\pi - 1)}$
 $\beta_\pi \uparrow \rightarrow$ slope of IS-MPR $\downarrow \rightarrow$ flatter.

α is larger
 $\pi_t^e \uparrow \rightarrow \pi_t \uparrow \rightarrow \pi_t > \pi_t^* \rightarrow i_t \downarrow \rightarrow r_t \uparrow \rightarrow COB \downarrow \rightarrow C_t \downarrow \rightarrow I_t \downarrow \rightarrow y_t \downarrow$
 α larger

INFLATION EXPECTATIONS AND INFLATION TARGET

1. We will start by repeating a graph from the last time.

[Figure 10: The IS-MPR-PC model when expected inflation equals the inflation target]

2. Figure 10 shows the case where both the temporary shocks $\epsilon_t^\pi = 0$ and $\epsilon_t^y = 0$ and the public's expectation of inflation equals the central bank's inflation target. $\pi^e = \pi_0 = \pi^*$
3. Specifically, the graph shows a case where the public's expectation of inflation and the central bank's inflation target are the same $\pi_t^e = \pi_0 = \pi^* = \pi_0$.

4. With no temporary shocks, the value of output is consistent with $\pi_t = \pi_0$ for the IS-MPR curve is y_t^* and $y_t = y_t^*$.

5. Now consider a case in which the public's inflation expectations shift to being higher than the central bank's target rate. Figure 11 illustrates this case. It shows the PC curve shifting upwards.

[Figure 11: The IS-MPR-PC model when expected inflation is higher]

6. This new PC is determined by the new higher level of expected inflation. Specifically, the public's inflation expectations are now determined by $\pi_t^e = \pi_1$.
7. The outcomes for inflation and output of the increase in inflation expectations are described by the intersection of the new PC and the old unchanged IS-MPR.
8. The actual outcome for inflation ends up being higher than the central bank's inflation target but lower than the public's inflationary expectations. Output ends up being lower than its natural rate because the higher level of inflation leads the central bank to raise real interest rates which reduces output.

9. Can We Learn More?

10. Figure 12 is a good example of how we can use graphs to illustrate a model's properties. It gets the basic story across as to what happens when inflation expectations rise above target when the central bank is pursuing a monetary policy rule that increases real rates in response to higher inflation.

11. Still, one could look to dig a bit deeper. The inflation outcome is slightly more than halfway towards the public's inflation expectations relative to the central bank's inflation target.

12. But what actually determines this outcome?

13. In other words, what determines how far from away target inflation will move when the public's inflation expectations change?

14. How much does it depend on the monetary policy rule? *BM*

15. How much does it depend on other aspects of the model, like the impact of real interest rates on output and the impact of output on inflation? *Y*

THE IS-MPR-PC MODEL SOLUTION FOR INFLATION

1. Let's repeat the equations describing our two curves as presented in our last set of notes. The PC curve is:

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \epsilon_t^\pi \quad (8)$$

2. And the IS-MP curve is:

$$y_t = y_t^* - \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \epsilon_t^y$$

3. Taking all the other elements of the model as given, we can view this as two linear equations in the two variables y_t and π_t . These equations can be solved to give solutions that describe how these two variables depend on all the other elements of the model.

4. Accordingly, we solve for π_t as:

$$\pi_t = \left(\frac{1 + \alpha\gamma(\beta_\pi - 1)}{1} \right) \pi_t^e + \left(\frac{\alpha\gamma(\beta_\pi - 1)}{1 + \alpha\gamma(\beta_\pi - 1)} \right) \pi_t^* + \frac{1 + \alpha\gamma(\beta_\pi - 1)}{\gamma\epsilon_y^t + \epsilon_\pi^t} \quad (9)$$

5. We define:

$$\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)}$$

so that we can re-write equation (2) for inflation as:

$$\pi_t = \theta\pi_t^e + (1 - \theta)\pi^* + \theta(\gamma\epsilon_y^t + \epsilon_\pi^t)$$

if $\theta = 0$ $\therefore \pi_t = \pi^$*

*average π_t^e & π^**

solu of π_t is the weighted

(10)

*if θ is small, then π_t is closer to π^**

$\alpha \downarrow, \gamma \downarrow, \beta \pi \downarrow$

6. This equation shows that, apart from the shocks to output and inflation ($\theta(\gamma\epsilon_y^t + \epsilon_\pi^t)$), inflation is a weighted average of the public's inflation expectations and the central bank's inflation target i.e. it must lie between these two values as long as $0 < \theta < 1$ (which it should be).

7. What determines whether inflation depends more on the public's expectations or the central bank's target? *π^**

θ

$$-\theta\alpha(\beta_\pi - 1) = \frac{1 + \alpha\gamma(\beta_\pi - 1)}{\alpha(\beta_\pi - 1)}$$

5. The coefficient determining this is $-\theta\alpha(\beta_\pi - 1)$. This can be re-expressed as:

the central bank's target?

4. How far does output fall short of its natural rate when inflation expectations rise above

$\Delta \pi_t^e > 0$

aggregate demand shocks ($\epsilon_t^d > 0$) have a positive effect on output.

(b) Inflationary supply shocks ($\epsilon_t^\pi > 0$) also have a negative effect on output while positive

having a negative effect on output.

(a) Increases in the public's inflation expectations relative to the inflation target end up

is negative. This means that:

3. Provided we have the usual condition that $\beta_\pi > 1$, the combined coefficient $-\theta\alpha(\beta_\pi - 1)$

ϵ_t^d and ϵ_t^π .

2. This equation tells us that whether output is above or below target depends upon the gap between expected inflation and the inflation target as well as on the two temporary shocks

$$y_t = y_t^* - \theta\alpha(\beta_\pi - 1)(\pi_t^e - \pi^* + \epsilon_t^\pi) + (1 - \theta\alpha\gamma(\beta_\pi - 1))\epsilon_t^d \quad (12)$$

1. Next we provide an expression for output and simplify the solution to output as:

THE IS-MPR-PC MODEL SOLUTION FOR OUTPUT

inflation and more weight on the central bank's inflation target.

bigger means less weight on in inflation expectations in determining the outcome for

reacting more to inflation being above its target level. So this parameter getting

(c) β_π : Let's continue to assume $\beta_\pi > 1$. Then as β_π gets bigger, the central bank is

inflation target will have more influence on the outcome for inflation.

rates. If the effect of interest rates on output gets bigger, then the central bank's

(b) α : This is the parameter that determines how output changes when output interest

influence on outcome for inflation.

output on inflation gets bigger, then the central bank's inflation target will have more

The central bank can only influence inflation by influencing output. If the effect of

(a) γ : This is the parameter that determines how inflation changes when output changes.

9. Three different factors determine this value:

8. In other words, what determines the value of θ ?

6. Calculating partial derivatives, you find that the size of the short fall in output depends positively on α and β_π .
7. In other words, the larger the impact of interest rates on output and the larger the central bank's interest rate response to inflation, the larger the shortfall in output will be when inflation expectations rise above the central bank's target.
8. In contrast, the output shortfall depends negatively on γ , the parameter determining the effect of output on inflation: As γ gets bigger, the central bank requires a smaller shortfall in output to implement its policy of getting inflation back to target.

HOW DO INFLATION EXPECTATIONS CHANGE?

1. Let's refer back to Figure 2. We have seen that after the public's inflation expectations rise, the result is a fall in output below its natural rate and in increase inflation, though this increase is smaller than had been expected by the public.

2. What happens next? How does the public's expectation of inflation change at this point?
3. Friedman's 1968 paper *The Role of Monetary Policy* suggested that people gradually adapt their expectations based on past outcomes for inflation.

4. Consider now a simple model of this idea of adaptive expectations by assuming that, each period, the expected level of inflation is simply equal to the level that prevailed last period. Formally, this can be written as:

$$(1) \quad \pi_t^e = \pi_{t-1}$$

$$PC: \pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t \pi$$

5. Under this formulation of expectations, the Phillips curve becomes:

$$\pi_t = \pi_{t-1} + \gamma(y_t - y_t^*) + \varepsilon_t \pi$$

$$(2) \quad \pi_t - \pi_{t-1} = \gamma(y_t - y_t^*) + \varepsilon_t \pi$$

6. By rearranging equation (2), we get:

$$\pi_t - \pi_{t-1} = \gamma(y_t - y_t^*) + \varepsilon_t \pi$$

$$\text{change in inflation} = \gamma(y_t - y_t^*) + \varepsilon_t \pi$$

$$(3) \quad \pi_t^e - \pi_{t-1} = \gamma(y_t - y_t^*) + \varepsilon_t \pi$$

7. In other words, there should be a **positive** relationship between the change in inflation and the output gap.

8. There are various methods for measuring output gaps but one quick and easy method is to use the unemployment rate as an indication of what the output might be.

- (a) If unemployment is high, then output is likely to be below its natural rate so the output gap is negative.

3. Figure 4 shows what happens after this. The PC curve moves back downwards and inflation moves down to a lower level, denoted on this graph by π_3 .

[Figure 4: Inflation expectations adjusting back downwards]

2. If people follow adaptive expectations then the next period, they will set $\pi^e = \pi_2$. (which is the central bank's inflation target) to π_2 .
1. After inflation expectations moved up to $\bar{\pi}$, the outcome was that inflation moved from π_1

ADJUSTMENT OF INFLATION EXPECTATIONS

14. Let's examine for now, however, how the IS-MP-PC model behaves when people have adaptive inflationary expectations.
13. That said, people are unlikely to simply use mechanical formulae to arrive at their expectations and one can imagine conditions in which people's inflation expectations could radically depart from what had happened in the past e.g. the appointment of a new central bank governor with a different approach to inflation, the adoption of a new currency or other major events.
12. These results suggest that the adaptive expectations approach appears to provide a reasonable model of how people formulate inflation expectations.
11. In contrast to the basic Phillips curve, this adaptive-expectations-style Phillips curve shows a clear and strong **negative** relationship between the change in inflation and the unemployment rate.
10. Figure 3 shows a scatter plot of data points on the change in inflation (measured as the four quarter percentage change in the price level minus the percentage change in the price level over the preceding four quarters) and the unemployment rate.

[Figure 3: Changes in inflation and unemployment, 1955-2013]

9. So if the adaptive expectations formulation of the Phillips curve was correct, then we would expect to see a **negative** relationship in the data between the change in inflation and the unemployment rate.
- (b) In contrast, a low unemployment rate is an indicator that the output gap is likely to be positive.

4. If the public has adaptive expectations, then inflation and output gradually converge back to the point where output is at its natural rate and inflation equals the central bank's target rate.
5. If the public has adaptive expectations, how could inflation expectations just jump upwards? Rather than a random unexplained increase in inflation expectations, the more likely explanation for the Phillips curve shifting upwards because is temporary supply shocks, i.e. ϵ_t^π is positive for a number of periods.
6. Under adaptive expectations, the public becomes used to higher inflation and so the Phillips curve will remain above its long-run position even after the temporary supply shock has been reversed.

A TEMPORARY AGGREGATE DEMAND SHOCK

1. Having looked at what happens under adaptive expectations when the Phillips curve shifts, let's consider what happens when we have a temporary shock to aggregate demand, so ϵ_t^d takes a different value from zero, which means a shift in the IS-MPR curve.
2. We will illustrate a case where there is a shift towards a positive value of ϵ_t^d for a couple of periods but then it shifts back to zero.

[Figure 5: A temporary aggregate demand shock ($\epsilon_t^d > 0$)]

3. Figure 5 shows the immediate impact of a positive aggregate demand shock. Output and inflation both go up with inflation reaching the point denoted as π_2 . If the public has adaptive expectations, then this results in an increase in inflation expectations the following period.
4. When the aggregate demand shock persists but inflation expectations move up to match the previous period's inflation rate, the inflation rate now rises again.

[Figure 6: Reversal of aggregate demand shock leads to recession with high inflation]

5. Figure 6 shows what happens if the aggregate demand shock then reverses itself in the next period. The IS-MPR curve shifts back to its original position but the Phillips curve remains elevated. The result is a nasty combination of high inflation and output below its natural rate:

- (a) An increase in output and inflation as the shock hits.
- (b) A further increase in inflation as inflation expectations adjust upwards, accompanied by a decline in output.
- (c) A decline in output and inflation as the shock disappears.
- (d) A further decline in inflation accompanied by an increase in output as inflationary expectations gradually return to the central bank's target.

6. This shows that when the public has adaptive inflation expectations, temporary positive aggregate demand shocks lead to counter-clockwise loops.

7. It turns out that much of the data on inflation and output correspond to these kinds of movements.

Figure 7 is borrowed from notes on Stanford economist Charles I. Jones's website. They show the data from US on inflation and an estimated output gap from 1960 to 1983. The figure shows a number of periods of clear counter-cyclical movements.

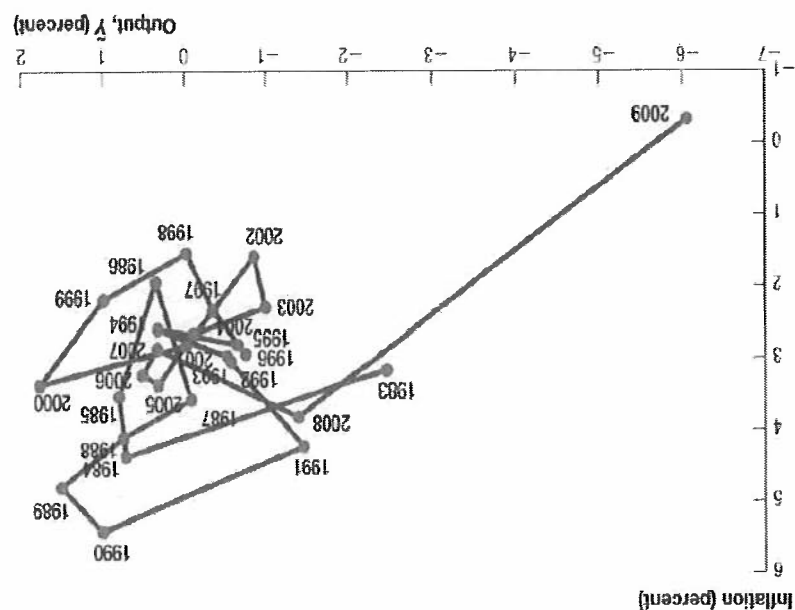


Figure 1: From Chad Jones: US inflation-output loops, 1960-1983

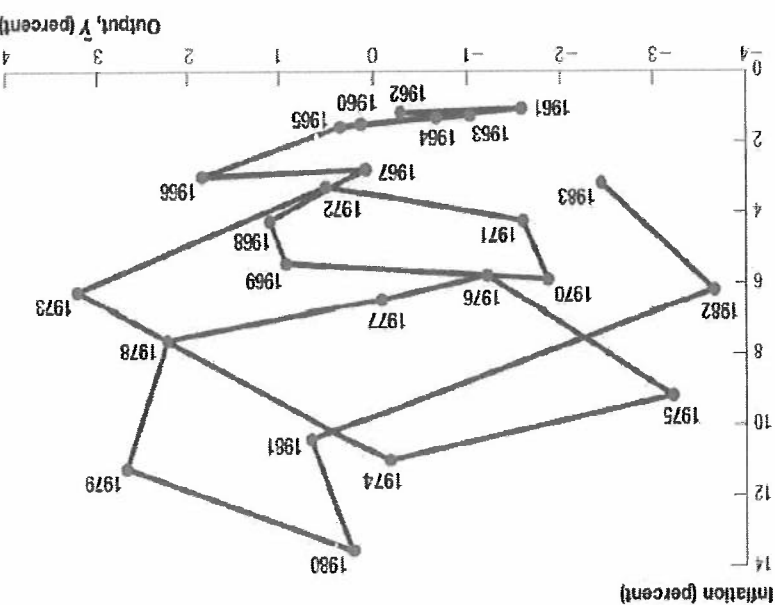


Figure 2: From Chad Jones: US inflation-output loops, 1983-2009

Figure 8 shows the same data from 1983-2009. This figure also shows some evidence counter-cyclical loops, though the movements are smaller than for the pre-1983 period.

WHAT IF INFLATION EXPECTATIONS DON'T ADJUST?

1. The evidence suggests that adaptive expectations seems to be a reasonable model for how people have formulated their expectations of inflation.
2. And it can be argued that it is a fairly convincing model of how people behave: Most people don't have the time or knowledge to fully understand exactly what's going in the economy and anticipating that last year's conditions provide a guide to what will happen this year probably works well enough for most people.

3. Indeed, if the value of θ is relatively high, then inflation will only change slowly under adaptive expectations, making the adaptive expectations assumption more accurate.
4. All that said, it is also possible to imagine situations in which the public's inflation expectations are not formed adaptively.

5. For example, if the public believes that the central bank will always act to return inflation quickly towards its target, then they may assume that deviations from the target will be temporary.

[Figure 9: A temporary positive demand shock when inflation expectations don't change]

6. Figure 9 shows how the economy reacts to a temporary positive demand shock when inflation expectations don't change.

7. The outcome here is much nicer than the counter-clockwise cycle. There is no recession at any point, just a short period of output being above its natural rate and inflation being above its target, followed by a return to their starting levels.

IMPLICATIONS FOR MONETARY POLICY

1. The previous examples provide food for thought about what kind of monetary policy we would like a central bank to implement.
2. Do we want a **soft** central bank that limits the increase in real interest rates when inflation rises to protect the economy and which is not too concerned about getting inflation back to target quickly?
3. Or do we want a **tough** central bank that raises interest rates aggressively and is very concerned about getting inflation back to target?

central banker who is more inflation-averse than they are.

(b) **Conservative central bankers:** If the central banker is known to really dislike inflation and the public believes this, the economy gets closer to the ideal low inflation outcome even without commitment. So the government may choose to appoint a

way to reassure the public about the bank's credibility.

(a) **Political independence:** A central bank that plans for the long-term and does not worry about economic performance during election years is more likely to stick to a commitment to low inflation. So, independence from political control is an important

- number of different ways:
2. How can this outcome be achieved? The academic literature on this topic has suggested a
1. These results predict that we get better outcomes if we have a **tough** central bank that acts aggressively against inflation and which the public believes is committed to keeping the economy near its inflation target.

IMPLICATIONS FOR CENTRAL BANK INSTITUTIONAL DESIGN

shock.

shows that happens when inflation expectations do not change after an expansionary comparing Figure 6 where inflation expectations shifts over time and Figure 9 which the less likely the economy is to go through boom-bust cycles. We can see this by (b) The more people believe that a central bank is maintaining its low inflation target, aggressive central bank.

if the original changes were smaller. So recessions are smaller and shorter with an natural level of output and the central bank's target inflation rate will also be faster smaller fall in output, i.e. a small recession. The process of a gradual return to the expectations. Equation (5) then tells us this more aggressive approach means a higher β_π is the smaller is the increase in inflation in response to a rise in inflation together tell us that the more aggressively the central bank responds to inflation (the in inflation. We can see this from equations (4) and (5). Equations (3) and (4) (a) The economy responds better to shocks if the central bank reacts strongly to changes

6. We can see this in two results from the model:

- its target rate.
5. However, our model says the exact opposite: Recessions are smaller and the economy less volatile when the central bank acts aggressively and is committed to returning inflation to
4. Suppose we wish to avoid large recessions if possible. You might imagine the soft central bank would be more likely to deliver this.

(c) **Consequence for bad inflation outcomes:** Introducing laws so that bad things happen to the central bankers when inflation is high is one way to make the public believe the they will commit to a low inflation rate.

Cramer's rule

$$a_{11}x + a_{12}y = c_1$$

$$a_{21}x + a_{22}y = c_2$$

This system can be written as:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

To solve for x

$$x = \frac{\begin{vmatrix} c_1 & a_{12} \\ c_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}} = \frac{c_1 a_{22} - c_2 a_{12}}{a_{11} a_{22} - a_{21} a_{12}}$$

$$y = \frac{\begin{vmatrix} a_{11} & c_1 \\ a_{21} & c_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}} = \frac{a_{11} c_2 - a_{21} c_1}{a_{11} a_{22} - a_{21} a_{12}}$$

(y) π_t

(x) y_t

PC

$$\pi_t = \pi_t^e + r(y_t - y_t^*) + \varepsilon_t \pi$$

IS-MPR $y_t - y_t^* = -\alpha(\beta\pi - 1)(\pi_t - \pi^*) + \varepsilon_t y$

We can rewrite these into

$$-r(y_t - y_t^*) + \pi_t = \pi_t^e + \varepsilon_t \pi$$

$$(y_t - y_t^*) + \alpha(\beta\pi - 1)\pi_t = \alpha(\beta\pi - 1)\pi^* + \varepsilon_t y$$

These can be rewritten as

$$\begin{bmatrix} -r & 1 \\ 1 & \alpha(\beta\pi - 1) \end{bmatrix} \begin{bmatrix} \Delta y_t - \Delta y_t^* \\ \Delta \pi_t \end{bmatrix} = \begin{bmatrix} \Delta \pi_t^e + \varepsilon_t \pi \\ \alpha(\beta\pi - 1)\Delta \pi^* + \Delta \varepsilon_t y \end{bmatrix}$$

$$y_t - y_t^* = \frac{\begin{vmatrix} \pi_t^e + \varepsilon_t \pi & 1 \\ \alpha(\beta\pi - 1)\pi^* + \varepsilon_t y & \alpha(\beta\pi - 1) \end{vmatrix}}{1 - r\alpha(\beta\pi - 1) - 1}$$

$$\pi_t = \frac{\begin{vmatrix} -r & 1 \\ \pi_t^e + \varepsilon_t \pi & \alpha(\beta\pi - 1)\pi^* + \varepsilon_t y \end{vmatrix}}{-r\alpha(\beta\pi - 1) - 1}$$

$$\varepsilon_t y \downarrow \rightarrow \Delta \varepsilon_t y > 0 \rightarrow \Delta y_t \uparrow \quad \Delta \pi_t \uparrow \quad \Delta \varepsilon_t y$$

$$\begin{bmatrix} -r & 1 \\ 1 & \alpha(\beta\pi - 1) \end{bmatrix} \begin{bmatrix} \Delta y_t \\ \Delta \pi_t \end{bmatrix} = \begin{bmatrix} \Delta \pi_t^e \\ 0 \end{bmatrix} + \begin{bmatrix} \Delta \varepsilon_t y \\ 0 \end{bmatrix}$$

$$\Delta y_t =$$

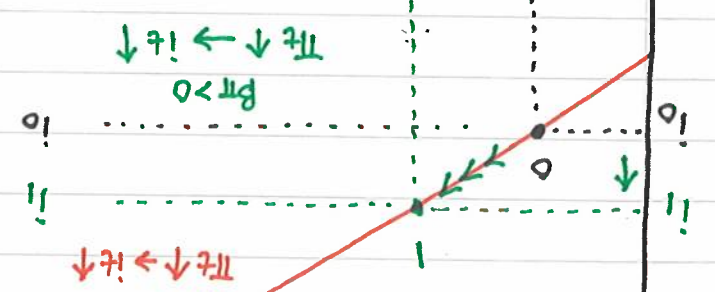
$$\Delta \pi_t = \frac{\begin{vmatrix} -r & 1 \\ 0 & \Delta \varepsilon_t y \end{vmatrix}}{-r\alpha(\beta\pi - 1) - 1} = \frac{-\Delta \varepsilon_t y}{-r\alpha(\beta\pi - 1) - 1} \cdot \Delta \varepsilon_t y > 0$$

$$\varepsilon_t y \downarrow \rightarrow \Delta \pi_t > 0 \rightarrow \pi_t \uparrow$$

Derivation of IS-MPR

$$i_t = r^* + \pi_t^* + \beta \pi_t (\pi_t - \pi^*)$$

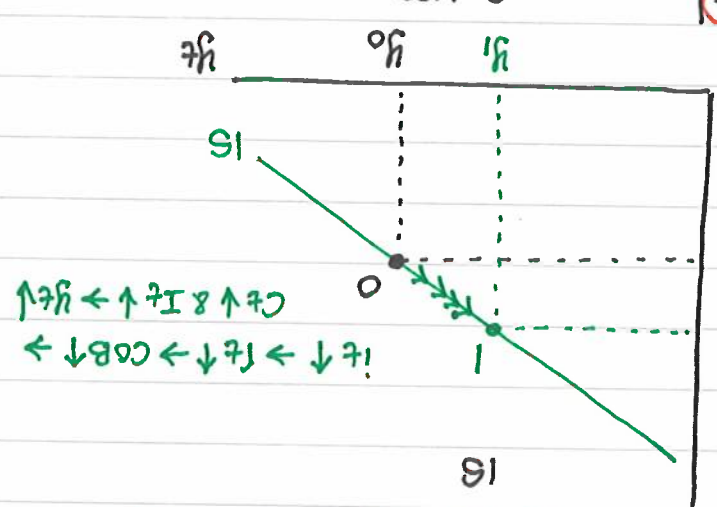
MPR i_t, r_t



$\pi_0 \rightarrow \pi_1$

$\pi_t \downarrow \rightarrow i_t \downarrow$

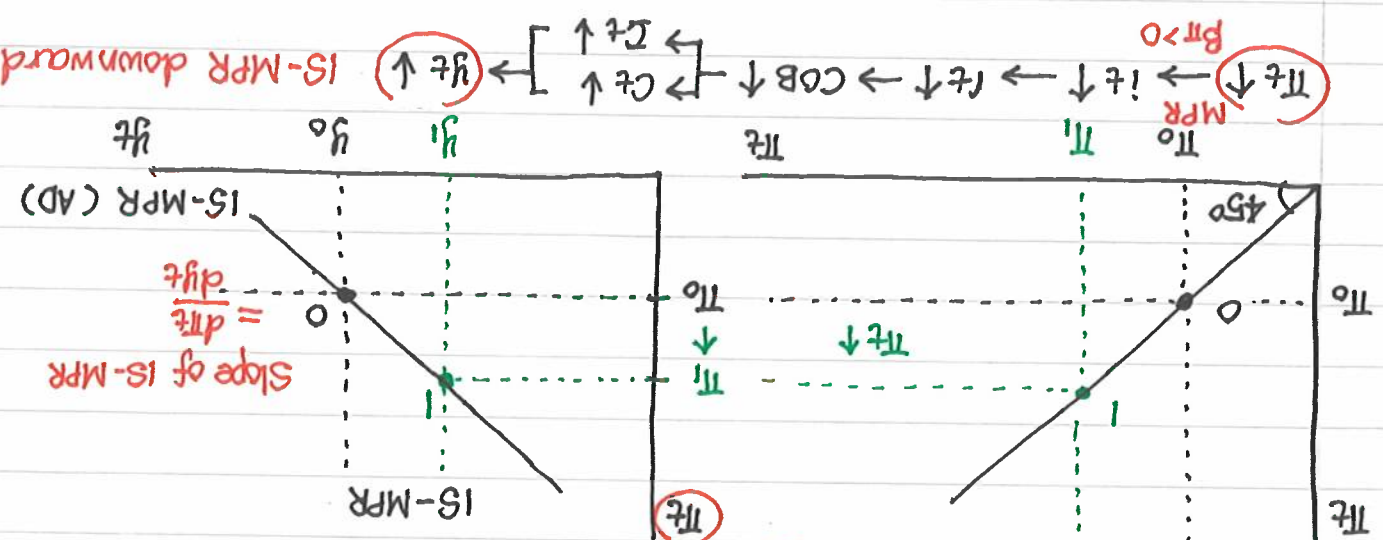
$\pi_t \uparrow \rightarrow i_t \uparrow$



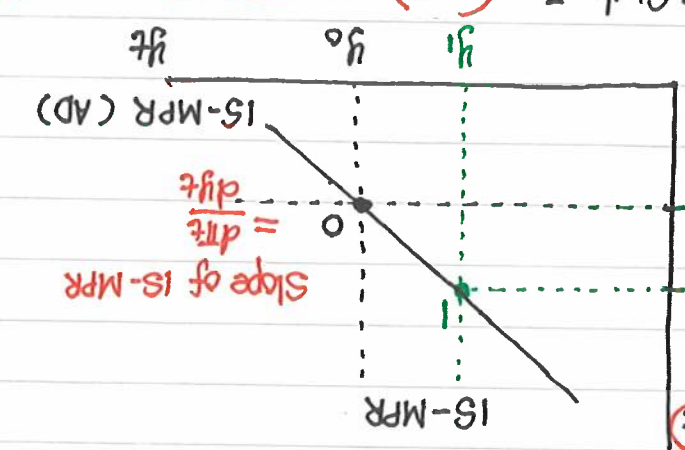
$$y_t = y_t^* - \alpha (r_t - r^*) + \varepsilon_t^y$$

$\downarrow i_t - \pi_t$

$i_t \downarrow \rightarrow r_t \downarrow \rightarrow C \downarrow \& I \downarrow \rightarrow y_t \downarrow$



$\pi_t \downarrow \rightarrow i_t \downarrow \rightarrow C \downarrow \& I \downarrow \rightarrow y_t \downarrow$



IS-MPR downward